

About This Document

The Infosys SP/DSE online test includes a Mathematical (Quantitative) Ability section of roughly 10-15 questions to be solved in about 25-35 minutes.

This document covers the recurring quantitative topics with practice questions and highlighted correct answers. Focus on speed and accuracy.

Number System

8 practice questions. The correct option is highlighted in green and also stated below each question.

Q1. Find the unit digit of 7^{95} .

(A) 1

(B) 3 ✓

(C) 7

(D) 9

Answer: (B) 3

Q2. Find the largest number that divides 62, 132 and 237 leaving the same remainder in each case.

(A) 30

(B) 35 ✓

(C) 42

(D) 70

Answer: (B) 35

Q3. How many trailing zeroes are there in $100!$?

(A) 20

(B) 24 ✓

(C) 25

(D) 10

Answer: (B) 24

Q4. Find the LCM of 12, 15 and 20.

(A) 60 ✓

(B) 120

(C) 180

(D) 30

Answer: (A) 60

Q5. What is the remainder when 2^{31} is divided by 5?

(A) 1

(B) 2

(C) 3 ✓

(D) 4

Answer: (C) 3

Q6. A number divided by 899 leaves remainder 63. What is the remainder when the same number is divided by 29?

(A) 5 ✓

(B) 4

(C) 3

(D) 63

Answer: (A) 5

Q7. Find the sum of the first 50 natural numbers.

(A) 1250

(B) 1275 ✓

(C) 1225

(D) 2550

Answer: (B) 1275

Q8. Find the HCF of 108 and 144.

(A) 12

(B) 24

(C) 36 ✓

(D) 18

Answer: (C) 36

Percentages

8 practice questions. The correct option is highlighted in green and also stated below each question.

Q1. Find 40% of 60% of 500.

(A) 100

(B) 120 ✓

(C) 150

(D) 200

Answer: (B) 120

Q2. If the price of a commodity increases by 25%, by what percent must consumption be reduced to keep expenditure unchanged?

(A) 25%

(B) 20% ✓

(C) 16.66%

(D) 10%

Answer: (B) 20%

Q3. 30% of a number is 135. Find the number.

(A) 405

(B) 450 ✓

(C) 500

(D) 350

Answer: (B) 450

Q4. In an exam 35% failed in Maths, 45% failed in English and 20% failed in both. What percent passed in both subjects?

(A) 40% ✓

- (B) 20%
- (C) 60%
- (D) 35%

Answer: (A) 40%

Q5. If 15% of x equals 20% of y, find x : y.

(A) 3 : 4

(B) 4 : 3 ✓

- (C) 2 : 3
- (D) 5 : 4

Answer: (B) 4 : 3

Q6. A number is increased by 20% and then decreased by 20%. Net change is:

- (A) No change
- (B) 4% increase

(C) 4% decrease ✓

(D) 2% decrease

Answer: (C) 4% decrease

Q7. A town's population is 10000 and grows 10% per year. Population after 2 years?

(A) 12000

(B) 12100 ✓

- (C) 11000
- (D) 12200

Answer: (B) 12100

Q8. A's salary is 20% more than B's. By what percent is B's salary less than A's?

- (A) 20%
- (B) 25%

(C) 16.66% ✓

(D) 18%

Answer: (C) 16.66%

Profit and Loss

8 practice questions. The correct option is highlighted in green and also stated below each question.

Q1. Cost price is Rs. 400 and selling price is Rs. 500. Find the profit percent.

(A) 20%

(B) 25% ✓

- (C) 10%
- (D) 15%

Answer: (B) 25%

Q2. An article sold at a 20% loss fetched Rs. 480. Find the cost price.

(A) Rs. 600 ✓

(B) Rs. 576

(C) Rs. 500

(D) Rs. 620

Answer: (A) Rs. 600

Q3. Marked price is 25% above cost price and a 10% discount is given. Find the profit percent.

(A) 15%

(B) 12.5% ✓

(C) 10%

(D) 13.5%

Answer: (B) 12.5%

Q4. If the cost price of 20 articles equals the selling price of 16 articles, find the profit percent.

(A) 20%

(B) 25% ✓

(C) 16%

(D) 30%

Answer: (B) 25%

Q5. A dishonest shopkeeper uses an 800 g weight for 1 kg. Find the profit percent.

(A) 20%

(B) 25% ✓

(C) 12.5%

(D) 16.66%

Answer: (B) 25%

Q6. Two articles are each sold for Rs. 990; one at 10% profit and the other at 10% loss. Net result is:

(A) No profit no loss

(B) Loss of Rs. 20 ✓

(C) Profit of Rs. 20

(D) Loss of Rs. 10

Answer: (B) Loss of Rs. 20

Q7. Selling price is Rs. 1140 at a profit of 14%. Find the cost price.

(A) Rs. 1000 ✓

(B) Rs. 1026

(C) Rs. 980

(D) Rs. 1100

Answer: (A) Rs. 1000

Q8. A profit of 20% is made on an item costing Rs. 250. Find the selling price.

(A) Rs. 280

(B) Rs. 300 ✓

(C) Rs. 320

(D) Rs. 270

Answer: (B) Rs. 300

Time and Work

8 practice questions. The correct option is highlighted in green and also stated below each question.

Q1. A completes a work in 10 days and B in 15 days. Working together, in how many days will they finish?

(A) 5

(B) 6 ✓

(C) 7.5

(D) 8

Answer: (B) 6

Q2. A is twice as efficient as B and together they finish a work in 12 days. In how many days can A alone finish it?

(A) 18 ✓

(B) 24

(C) 16

(D) 20

Answer: (A) 18

Q3. If 10 men finish a job in 8 days, how long will 8 men take for the same job?

(A) 10 days ✓

(B) 6 days

(C) 12 days

(D) 9 days

Answer: (A) 10 days

Q4. Pipe A fills a tank in 6 hours and pipe B in 4 hours. Together they fill it in:

(A) 2 hours

(B) 2.4 hours ✓

(C) 3 hours

(D) 2.5 hours

Answer: (B) 2.4 hours

Q5. A and B together do a work in 8 days, B and C in 12 days and A, B and C together in 6 days. C alone takes:

(A) 24 days ✓

(B) 12 days

(C) 20 days

(D) 16 days

Answer: (A) 24 days

Q6. A cistern is filled by tap A in 4 hours and emptied by tap B in 9 hours. If both are open, time to fill:

(A) 6 hours

(B) 7.2 hours ✓

(C) 8 hours

(D) 5 hours

Answer: (B) 7.2 hours

Q7. 15 workers finish a job in 40 days. After 10 days, 5 workers leave. How many more days to finish?

(A) 45 days ✓

(B) 40 days

(C) 50 days

(D) 30 days

Answer: (A) 45 days

Q8. A can do a work in 20 days and B in 30 days. They start together but A leaves after 5 days. Total days to finish:

(A) 22.5 ✓

(B) 20

(C) 25

(D) 17.5

Answer: (A) 22.5

Time, Speed and Distance

8 practice questions. The correct option is highlighted in green and also stated below each question.

Q1. Convert 90 km/h into m/s.

(A) 25 m/s ✓

(B) 30 m/s

(C) 20 m/s

(D) 15 m/s

Answer: (A) 25 m/s

Q2. A 150 m long train running at 90 km/h crosses a pole in:

(A) 6 s ✓

(B) 8 s

(C) 10 s

(D) 5 s

Answer: (A) 6 s

Q3. A car covers a distance at 40 km/h and returns at 60 km/h. Find the average speed for the whole journey.

(A) 50 km/h

(B) 48 km/h ✓

(C) 45 km/h

(D) 52 km/h

Answer: (B) 48 km/h

Q4. Two trains of lengths 140 m and 160 m run in opposite directions at 60 km/h and 40 km/h. Time to cross each other:

(A) 10.8 s ✓

(B) 12 s

(C) 9 s

(D) 15 s

Answer: (A) 10.8 s

Q5. A boat's speed in still water is 10 km/h and the stream is 2 km/h. Downstream speed is:

(A) 8 km/h

(B) 12 km/h ✓

(C) 10 km/h

(D) 6 km/h

Answer: (B) 12 km/h

Q6. Walking at $\frac{5}{6}$ of his usual speed a man is 10 minutes late. His usual time is:

(A) 40 min

(B) 50 min ✓

(C) 60 min

(D) 45 min

Answer: (B) 50 min

Q7. A distance of 120 km is covered. If speed increases by 10 km/h, 1 hour is saved. Original speed is:

(A) 25 km/h

(B) 30 km/h ✓

(C) 40 km/h

(D) 20 km/h

Answer: (B) 30 km/h

Q8. A man covers 300 km in 5 hours. His speed is:

(A) 50 km/h

(B) 60 km/h ✓

(C) 65 km/h

(D) 55 km/h

Answer: (B) 60 km/h

Ratio and Proportion

8 practice questions. The correct option is highlighted in green and also stated below each question.

Q1. Divide Rs. 600 in the ratio 2 : 3. The smaller share is:

(A) Rs. 200

(B) Rs. 240 ✓

(C) Rs. 260

(D) Rs. 300

Answer: (B) Rs. 240

Q2. If $A : B = 2 : 3$ and $B : C = 4 : 5$, find $A : B : C$.

(A) 8 : 12 : 15 ✓

(B) 2 : 3 : 5

(C) 8 : 12 : 10

(D) 2 : 4 : 5

Answer: (A) 8 : 12 : 15

Q3. Find the mean proportional between 4 and 16.

(A) 6

(B) 8 ✓

(C) 10

(D) 12

Answer: (B) 8

Q4. Ratio of two numbers is 5 : 8. If each is increased by 4 the ratio becomes 2 : 3. The numbers are:

(A) 20 and 32 ✓

(B) 15 and 24

(C) 25 and 40

(D) 10 and 16

Answer: (A) 20 and 32

Q5. Two numbers are in the ratio 3 : 5 and their sum is 64. The larger number is:

(A) 24

(B) 40 ✓

(C) 35

(D) 45

Answer: (B) 40

Q6. If $a/b = 2/3$ and $b/c = 4/5$, find $a : c$.

(A) 8 : 15 ✓

(B) 2 : 5

(C) 8 : 12

(D) 6 : 15

Answer: (A) 8 : 15

Q7. Divide Rs. 1200 among A, B and C in the ratio 3 : 4 : 5. C's share is:

(A) Rs. 400

(B) Rs. 500 ✓

(C) Rs. 300

(D) Rs. 480

Answer: (B) Rs. 500

Q8. If $3 : 4 = x : 20$, then x equals:

(A) 12

(B) 15 ✓

(C) 18

(D) 16

Answer: (B) 15

Averages

8 practice questions. The correct option is highlighted in green and also stated below each question.

Q1. Find the average of the first 10 natural numbers.

(A) 5

(B) 5.5 ✓

(C) 6

(D) 4.5

Answer: (B) 5.5

Q2. The average of 5 numbers is 27. If one number is 45, the average of the remaining four is:

(A) 22.5 ✓

(B) 20

(C) 24

(D) 25

Answer: (A) 22.5

Q3. The average age of 30 students is 12 years. Including the teacher the average rises by 1. The teacher's age is:

(A) 41

(B) 43 ✓

(C) 45

(D) 40

Answer: (B) 43

Q4. A cricketer has an average of 32 in 10 innings. How many runs must he score in the 11th innings to raise his average to 34?

(A) 54 ✓

(B) 52

(C) 56

(D) 50

Answer: (A) 54

Q5. Average weight of A, B and C is 45 kg. A and B average 40 kg, B and C average 43 kg. Weight of B is:

(A) 31 kg ✓

(B) 33 kg

(C) 29 kg

(D) 35 kg

Answer: (A) 31 kg

Q6. Find the average of all even numbers from 2 to 20 (inclusive).

(A) 10

(B) 11 ✓

(C) 12

(D) 9

Answer: (B) 11

Q7. The average of 50 numbers is 30. If two numbers 35 and 40 are removed, the new average (to 2 dp) is:

(A) 29.69 ✓

(B) 30.10

(C) 28.50

(D) 29.00

Answer: (A) 29.69

Q8. Find the average of 20, 40 and 60.

(A) 35

(B) 40 ✓

(C) 45

(D) 50

Answer: (B) 40

Simple and Compound Interest

8 practice questions. The correct option is highlighted in green and also stated below each question.

Q1. Find the simple interest on Rs. 5000 at 10% per annum for 2 years.

(A) Rs. 1000 ✓

(B) Rs. 1100

(C) Rs. 500

(D) Rs. 1050

Answer: (A) Rs. 1000

Q2. Find the compound interest on Rs. 10000 at 10% per annum for 2 years.

(A) Rs. 2000

(B) Rs. 2100 ✓

(C) Rs. 2200

(D) Rs. 1900

Answer: (B) Rs. 2100

Q3. A sum doubles itself in 8 years under simple interest. The annual rate is:

(A) 10%

(B) 12.5% ✓

(C) 15%

(D) 8%

Answer: (B) 12.5%

Q4. The difference between CI and SI on a sum at 10% per annum for 2 years is Rs. 50. The sum is:

(A) Rs. 5000 ✓

- (B) Rs. 4000
- (C) Rs. 6000
- (D) Rs. 4500

Answer: (A) Rs. 5000

Q5. Compound interest on Rs. 8000 at 10% per annum for 1 year, compounded half-yearly, is:

(A) Rs. 800

(B) Rs. 820 ✓

- (C) Rs. 840
- (D) Rs. 810

Answer: (B) Rs. 820

Q6. A sum amounts to Rs. 1210 in 2 years and Rs. 1331 in 3 years under compound interest. The rate is:

(A) 8%

(B) 10% ✓

- (C) 12%
- (D) 11%

Answer: (B) 10%

Q7. Find the amount on Rs. 8000 at 5% per annum simple interest for 3 years.

(A) Rs. 9000

(B) Rs. 9200 ✓

- (C) Rs. 9600
- (D) Rs. 8800

Answer: (B) Rs. 9200

Q8. In how many years will Rs. 1200 become Rs. 1600 at 8.33% per annum simple interest?

(A) 3 years

(B) 4 years ✓

- (C) 5 years
- (D) 6 years

Answer: (B) 4 years

Permutation, Combination and Probability

8 practice questions. The correct option is highlighted in green and also stated below each question.

Q1. What is the value of $5!$?

- (A) 60
- (B) 100

(C) 120 ✓

(D) 24

Answer: (C) 120

Q2. In how many ways can the letters of the word APPLE be arranged?

(A) 120

(B) 60 ✓

(C) 24

(D) 20

Answer: (B) 60

Q3. In how many ways can 3 items be chosen from 10 distinct items?

(A) 720

(B) 120 ✓

(C) 30

(D) 100

Answer: (B) 120

Q4. Two dice are rolled. What is the probability that the sum is 7?

(A) $\frac{1}{6}$ ✓

(B) $\frac{1}{8}$

(C) $\frac{1}{12}$

(D) $\frac{1}{9}$

Answer: (A) $\frac{1}{6}$

Q5. From 4 boys and 3 girls, in how many ways can a committee of 2 boys and 2 girls be formed?

(A) 12

(B) 18 ✓

(C) 24

(D) 36

Answer: (B) 18

Q6. A card is drawn from a standard deck. Probability that it is a king:

(A) $\frac{1}{13}$ ✓

(B) $\frac{1}{4}$

(C) $\frac{1}{52}$

(D) $\frac{4}{13}$

Answer: (A) $\frac{1}{13}$

Q7. In how many ways can 4 people be seated in a row?

(A) 12

(B) 16

(C) 24 ✓

(D) 8

Answer: (C) 24

Q8. A fair coin is tossed once. Probability of getting a head:

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$ ✓

(C) 1

(D) $\frac{1}{3}$

Answer: (B) $\frac{1}{2}$



Quantitative Aptitude

Applied Role(s):

Digital Specialist Engineer (Trainee) | Specialist Programmer L1 / L2 / L3 (Trainee)

Online Assessment Preparation Material

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Compiled from verified previous-year patterns, community-reported questions and research-backed company-style practice.

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